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TEC–401

B. TECH. (ECE)
(FOURTH SEMESTER)
MID SEMESTER
EXAMINATION, March, 2024

CONTROL SYSTEM

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each sub-question carries 10 marks.

1. (a) The concept of transfer functions is central to understanding the behavior of dynamic systems. Discuss in detail what a transfer function represents and how it is derived from a system's differential equations. Also define poles and zeros of a transfer function using pole zero plot. (CO1)

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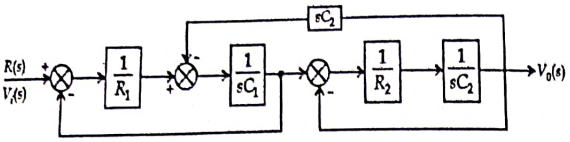
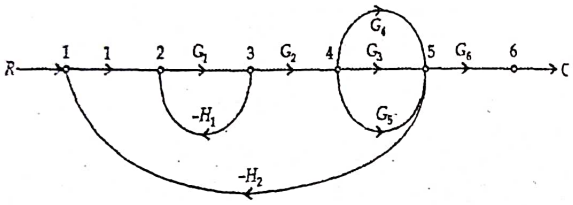
OR

(b) Obtain the transfer function of the feedback control system shown in the figure by the block diagram reduction technique : (CO1)



2. (a) Obtain the overall transfer function C/R from the signal flow graph shown in figure : (CO1)



OR

(b) Define open-loop and closed-loop control systems. Outline the derivation of the

closed-loop transfer function and highlight its importance in system analysis. Support your explanation with a brief real-world example. (CO1)

3. (a) Explain the Routh stability criterion. Investigate the stability : (CO2)

$$s^5 + s^4 + 2s^3 + 2s^2 + 3s + 5 = 0$$

OR

(b) When a second order control system is subjected to a unit step input, the values of $\zeta = 0.5$ and $w_n = 6$ rad/sec, determine the rise time, peak time, settling time and peak overshoot. (CO2)

4. (a) Explain how to derive the response of a first-order system to a unit step input briefly, covering system characteristics, the derivation process, the role of the time constant. (CO2)

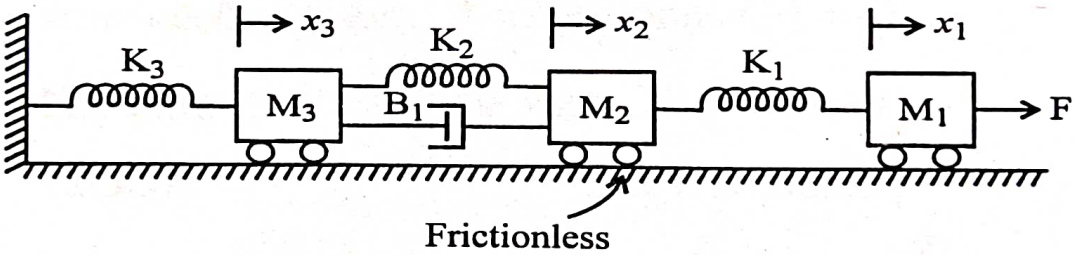
OR

(b) Explain the concept of percent maximum overshoot in control system response.

How is it calculated and what does it reveal about the system's behavior ?

(CO2)

5. (a) Derive the values of static error coefficients and steady state error for a type 1 system with unit step input. (CO2)

OR

(b) Draw the F-V analogy of the mechanical system shown in figure : (CO1)